

Effect of Bumble Bee (Hymenoptera: Apidae) Pollination Intensity on the Quality of Greenhouse Tomatoes

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ABSTRACT Laboratory studies were conducted to assess tomato, *Lycopersicon esculentum* Mill. (Solanaceae), quality in relation to the level of buzz-pollination by bumble bees. Studies were conducted in commercial tomato greenhouses in the Leamington, Ontario, area to categorize bruising of tomato anther cones by bumble bees into five levels of bruising. The number of pollen grains per stigma was determined for each bruising level, and the bruising level was found to be a good predictor of stigmatic pollen load. Experimental flowers were pollinated by bumble bees and assigned to bruising levels based on the degree of anther cone discoloration. Fruit set, tomato weight, minimum diameter, the number of days until ripe, roundness, weight, percentage sugars, and number of seeds were assessed and compared among bruising level. Fruit set in flowers receiving no pollination visits was 30.2%, whereas, 83.3, 84.4, 81.2, and 100% of the flowers set fruit in bruising levels 1, 2, 3, and 4, respectively. Minimum diameter, number of seeds, and tomato weight all increased from no bruising to different levels of bruising. There was no increase in weight or diameter above a bruising level of 1, and no increase in the number of seeds per fruit after a bruising level of 2. We found that pollination of tomato flowers greater than a bruising level of 2 (corresponding to approximately one to two bee visits) did not result in a significant increase in quality.

KEY WORDS bumble bee, tomato quality, greenhouse, pollination

ALTHOUGH THE TOMATO, *Lycopersicon esculentum* Mill. (Solanaceae), is largely self-fertile; the flowers require agitation of the anther cone to release pollen from the anthers (Buchmann 1983, 1986). In greenhouses, this has been achieved through the use of hand-held electric pollinators (Picken 1984). More recently, bumble bees have been found to be at least as effective, and in many cases more effective, than manual pollination or honey bee pollination in terms of the quantity and quality of tomatoes produced (Banda and Paxton 1991, Kevan et al. 1991, van Ravestijn and van der Sande 1991, Straver and Plowright 1991, Dogterom et al. 1998, Pressman et al. 1999). By 1994, all commercial tomato greenhouses in southern Ontario were using bumble bees as their sole means of pollination (S. Khosla, Ontario Ministry of Agriculture, Food and Rural Affairs, Harrow, ON, personal communication).

Although experiments with bumble bee pollination of tomato flowers has been found to produce high quality fruit in terms of yield, seed set, fruit weight, and roundness (Kevan et al. 1991, van Ravestijn and van der Sande 1991, Straver and Plowright 1991), there is little information on how different levels of pollination by bumble bees affects quality. Too few visits by bees may result in a low number of fertilized ovules, and substandard fruit (e.g., Verkerk 1957, van Rav-

estijn and van der Sande 1991, Straver and Plowright 1991, Pressman et al. 1998). Too much bee activity on a flower may cause damage to the reproductive organs, premature loss of the flower, and no fruit development (Cribb 1990).

Studies on various plant species such as *Turnera ulmifolia* L. (Shore and Barrett 1984), *Passiflora vitifolia* HBK (Snow 1982), and *Delphinium nelsonii* Greene (Waser and Price 1991) have found that as pollen dose increases, so does seed set. There is also a large body of literature showing positive correlations among the number of seeds per fruit, fruit weight, sugar content, and fruit shape (e.g., Hong et al. 1991, Chen 1996, Gonzalez et al. 1998, Suzuki et al. 1998).

Typically, more pollinator visits result in better quality fruit (Chagnon et al. 1989, Chen 1996, Pressman et al. 1998). Chen (1996) found that the percentage fruit set and quality of cantaloupe increased with greater numbers of pollination visits by honey bees. Similarly, Pressman et al. (1998) found that in sweet peppers, greater numbers of manual pollinations per flower resulted in larger, improved shape fruit. They concluded that an increase in the number of pollinations could greatly increase size and quality of yield.

Few studies have examined tomato quality in relation to levels of pollination. Fletcher and Gregg (1907) showed that, within limits, the size of a tomato is dependent on how much pollen is transferred to the stigma. Verkerk (1957) found that the more often tomato flowers were manually pollinated, up to a total of four times, the greater the percentage fruit set and the higher the number of seeds per fruit. He also found

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that a greater number of pollen grains on the stigma resulted in faster fruit development and therefore an earlier and more profitable crop.

The level of bumble bee activity on a tomato flower is straightforward to assess. Bin and Sorressi (1973) observed that pollination of tomato flowers by bumble bees caused bruising or necrotic spotting on the anther cone. In European greenhouses, van Ravestijn and van der Sande (1991) assessed the number of tomato flowers that had been pollinated by the bumble bee *Bombus terrestris* L. by observing the brown discoloration of the anther cone caused by pollination visits. Although some studies have examined fruit quality in relation to the number of pollination visits, total duration of pollination visits is a better measure because it is a more accurate indication of pollen transfer (Jarlan et al. 1997). Thus, the level of discoloration of tomato anther cones may be a good indicator of stigmatic pollen load because the intensity of discoloration corresponds to the duration of contact by bumble bees.

In southern Ontario, greenhouse growers and suppliers use necrotic discoloration of the anther cone as an estimate of the proportion of flowers visited, and as an indication of the amount of pollination within the greenhouse. Most suppliers and growers in Ontario want 80–100% of the flowers with heavy bruising on the anther cone (S. Khosla, Ontario Ministry of Agriculture, Food and Rural Affairs, Harrow, ON, personal communication). Currently, in southern Ontario, *Bombus impatiens* Cresson colonies are added to greenhouses largely based on this method of pollination assessment. Nonetheless, there has been no research to determine if visual evaluation of the degree of necrotic bruising on anther cones is a reliable method of assessing stigmatic pollen load, or what degree of bruising is required for optimal tomato yield and quality.

In this study, we addressed two questions: (1) Is the degree of necrotic discoloration on tomato anther cones a good predictor of stigmatic pollen load? (2) What is the relationship between the level of necrotic bruising on tomato flowers and various fruit characteristics? We predicted that measures of fruit quality would increase rapidly with slight bruising and then level off before anther cones were heavily bruised.

Materials and Methods

To determine the relationship between bruising and tomato quality, we first developed a rating scale to categorize the degree of bruising. We then assessed the relationship between the level of bruising on the anther cone, and the stigmatic pollen load. The relationship between bruising level and fruit characteristics was then evaluated.

Study Area and Organisms. The study took place in an experimental greenhouse at the University of Western Ontario from 1 March 1999–18 October 1999. The greenhouse was ≈ 6 m wide by 6 m long by 3 m high, and temperature was regulated using screened roof vents at 22°C in the day and 20°C at night. Approxi-

mately 100 tomato plants (*L. esculentum* 'Trust') were grown from seeds planted in sterilized potting soil. For the first month after the seeds were planted, they were given plant starter fertilizer (10–52–10, N-P-K, Plant-Prod, Brampton, ON) every week. For the rest of the experiment, the plants were given all purpose fertilizer (20–20–20, N-P-K, Plant-Prod), weekly. The plants were watered daily, or more often, if needed. The experiment began shortly after the first flowers appeared.

Colonies of *B. impatiens* were obtained from two bumble bee supply companies, Koppert Biological Systems Canada (Scarborough, ON), and Biobest Canada (Leamington, ON). Colonies from both suppliers were housed in plastic containers ≈ 20 by 28 by 18 cm, surrounded by an outer cardboard casing. Upon delivery, each colony contained ≈ 30 –60 workers and one queen. Colonies were used for no more than 2 mo, after which they began to produce male bees and new queens. Queens and males, or reproductives, do not forage for the colony, and, consequently, colonies at this stage are no longer useful for greenhouse pollination. Because tomato flowers do not produce nectar, a required food for bumble bees, colonies were equipped with a bag containing a nectar substitute. Foragers moved in and out of the hive through a 1.5-cm hole in the hive casing.

Bruising Levels. Five bruising categories (0–4) were designated based on visual assessment of the degree of necrotic discoloration on tomato flower anther cones as a result of buzz pollination by bumble bees (Fig. 1). The fifth bruising category, representing extremely heavy bruising and destruction of the anther cone, was added after the tomato quality experiment had begun so that no data were collected on number of pollen grains for this category. Bruising level 4 was never observed in the commercial greenhouses, but did occur occasionally in the small experimental greenhouse. Fig. 1 and Table 1 were used as guides in assigning tomato flowers to the bruising groups based on degree of necrotic discoloration. The typical number of bee visits resulting in bruising levels 1 to 4 was estimated by observing bee activity on flowers in commercial tomato greenhouses. Approximately five flowers from each bruising category were tagged and observed until visited by a bee. The flower was then covered with a mesh bag to prevent further bee visits. After 2 h had elapsed, the bruising level of the flower was reassessed. The typical number of bee visits recorded in Table 1 is an approximation of the number of bee visits associated with the bruising levels, but can vary depending on the length of the bee visit.

Flowers were examined under a microscope to determine if the level of bruising corresponded to the number of pollen grains on the stigma. Six flowers from each category were randomly selected, dissected immediately after being picked, and the stigmas removed. The pollen grains on the stigmas were then stained with basic fuchsin jelly (after Kearns and Inouye 1993). The number of pollen grains on each stigma was counted under a compound microscope.



Fig. 1. Bruising levels of tomato flowers based on necrotic discoloration of the anther cone as a result of buzz pollination by bumble bees. Bruising level 0: no pollination. Bruising level 1: lightly pollinated. Bruising level 2: moderately pollinated. Bruising level 3: heavily pollinated. Bruising level 4: very heavily pollinated.

Analysis of variance (ANOVA) (Minitab 1997) was used to determine if numbers of pollen grains were different among the first four bruising categories. To meet the ANOVA assumption of homogeneity of variances, the data were transformed using Taylor's power law (Taylor 1961). All reported averages are from the nontransformed data. Differences among treatment groups were assessed using Tukey's pairwise comparison tests. The association between the number of pollen grains per stigma and bruising level was examined using Spearman's correlation (Spearman 1904).

Tomato Quality. One colony was introduced into the experimental greenhouse and allowed to forage on the tomato flowers for 2–3 d. The hive's exit hole was

then covered, leaving only the entrance hole open. After all bees had returned to the colony, a randomly chosen subset of flowers were tagged. Each tag was labeled with a unique number, dated, and the level of bruising was assessed using the criteria outlined in Fig. 1 and Table 1. Bees were kept inside the colony for at least 4 d, when all of the tagged flowers had either completely closed or fallen off. The colony was then reopened, and the tagging and recording was repeated every 2–3 wk. Flowers were tagged on 11 d and the number of flowers tagged in each category was approximately the same on each day of tagging.

All tomatoes were harvested at the same ripeness, based on visual assessment of color. The tomatoes were considered ripe when they were a uniform orange-red color. The number of tomato flowers that set fruit, verses the number that did not set, was compared among bruising level groups using 4×2 chi-square contingency analysis ($df = 3$). Minimum and maximum tomato diameters were measured using digital callipers. Average diameter was calculated from the minimum and maximum diameters. Tomato roundness was calculated using a revised equation taken from Kevan et al. (1991) (equation 1). The revised equation took into account minimum and maximum diameter, but left out the calyx to stigmatic scar diameter because when tomatoes are assessed for roundness in greenhouses, it is the equatorial roundness that is important for efficient packaging (S. Khosla, Ontario Ministry of Agriculture, Food and Rural Affairs, Harrow, ON, personal communication).

$$\text{Roundness} = \sum_{i=1}^2 |1 - d_i/d_{\text{ave}}|, \quad [1]$$

where d_i = diameter

$$d_{\text{ave}} = \text{average diameter} = (d_1 + d_2)/2.$$

Tomatoes were weighed to 0.01 g using an electronic scale (Mettler PJ300, Columbus, OH). The entire tomato was then homogenized in a commercial food blender, and the percentage of sugars of the blended mixture was measured to one decimal place using a hand-held 0–50% refractometer (Bellingham and Stanley, Atlanta, GA). The blended mixture was then rinsed through a 1-mm sieve to separate the seeds from the liquid. The total seeds per tomato were then counted. The number of days until ripe was calculated starting from the date the flower was tagged, until the date the tomato was harvested.

The data were first analyzed by MANOVA (multivariate ANOVA, Minitab 1997), with roundness, weight, sugars, number of seeds, minimum diameter, the difference in diameter between the minimum and maximum diameter, and days until ripe as the response variables. MANOVA was followed by univariate ANOVA and Tukey's pairwise comparisons. The nine variables were compared using Spearman's correlation analysis.

The Canadian Food Inspection Agency grades and standards for greenhouse tomatoes (1999) categorizes tomatoes into three groups: Canada number 1 grade,

Table 1. Bruising level categories of tomato flowers based on levels of necrotic discoloration on the anther as a result of bumble bee (*Bombus impatiens*) buzz pollination

Bruising level	Visual assessment	Typical no. of bee visits
0	No bruising	0
1	One or two small areas of discoloration	1
2	Two to three small to medium size discolorations	1–2
3	One or more large, or greater than 3 medium discolorations	2–4
4	Entire anther cone bruised, and anthers coming apart	>4

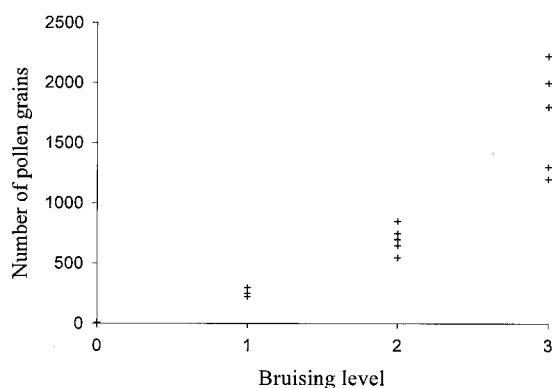


Fig. 2. Relationship between the number of pollen grains on the stigma of tomato flowers and the bruising level resulting from bruising of the anther cone from buzz pollination by bumble bees ($n = 24$).

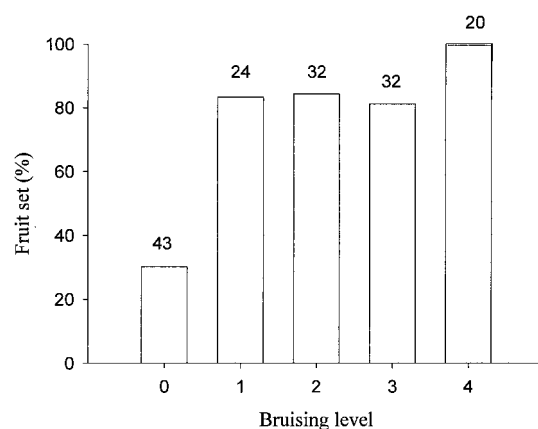


Fig. 3. Percent fruit set of tomato flowers in relation to the level of bumble bee pollination. Numbers above each bar indicate the sample size for each bruising level.

Canada commercial grade, and Canada number 2 grade. One of the main criteria that distinguish the Canada commercial grade from the Canada number 2 grade is a minimum diameter for the commercial grade of 63 mm. Based on these criteria, if the minimum diameters were ≥ 63 mm, we categorized the tomatoes as commercial grade. If the minimum diameters were < 63 mm, we categorized the tomatoes as number 2 grade. The Canadian standards outline additional factors, which we did not take into account, such as the amount of creasing and the extent of growth cracks, when tomatoes are graded. The number of tomatoes in each grade in each bruising level was compared using a 2×5 chi-square contingency table ($df = 4$). For all chi-square analyses, when data sets had expected values of < 5.0 , the data were adjusted using Yates' correction (Yates 1934).

Results

Bruising Levels. Bruising categories based on the degree of discoloration differed in the mean number of pollen grains per stigma ($r_s = 0.971$; $F = 168.05$; $df = 3, 20$; $P < 0.0001$). Tukey's pairwise comparison test showed that the mean number of pollen grains was different among the four groups ($P < 0.0001$ for all comparisons). There was a strong positive correlation between the number of pollen grains and the bruising level ($r_s = 0.97$, $n = 24$, $P < 0.001$; Fig. 2), indicating

that the degree of anther cone bruising is a good predictor of stigmatic pollen load.

Tomato Quality. The number of tomatoes that set versus the number of flowers that did not set differed among bruising level groups ($\chi^2 = 48.24$, $df = 4$, $P < 0.0001$). The bruising treatment group contributing most to the difference was level 0, no pollination, with 13 set tomatoes, and 30 flowers that did not set ($\chi^2 = 32.83$). With 30.2% of the flowers that set, bruising level 0 had the lowest percentage of fruit set. Bruising level 1 had a fruit set of 83.3%, level 2 had 84.37%, level 3 had 81.25%, and level 4 had the highest with 100% fruit set (Fig. 3). When bruising level 0 was removed from the analysis, there was no difference in the number of flowers that set to the number that did not set ($\chi^2 = 2.47$, $df = 3$, $P = 0.498$). This suggests that a greater degree of pollination did not increase fruit set over levels 1–4. When bruising levels 1–3 were pooled and compared with level 4, there was no difference in fruit set ($\chi^2 = 2.67$, $df = 1$, $P = 0.113$).

Multivariate ANOVA on the five bruising levels showed a difference among bruising treatments with respect to roundness, weight, sugars, number of seeds, average diameter, difference in diameter between the minimum and maximum diameter, and minimum diameter and days until ripe ($F = 2.60$; $df = 28, 384$; $P < 0.0001$; $1 - \beta = 1.00$). Univariate ANOVA showed that there was a difference in the average diameter ($F = 4.31$; $df = 4, 99$; $P = 0.003$; $1 - \beta = 0.920$), minimum

Table 2. Comparison of seven measures of tomato quality recorded from four bruising level treatment groups

Bruising level	Diam difference, mm	Min. diam, mm	Days to ripen	Roundness	No. of seeds	Sugars, %	Weight, g
0	1.52 $\pm$ 1.34a	56.37 $\pm$ 2.18a	68.69 $\pm$ 3.64a	0.026 $\pm$ 0.02a	71.69 $\pm$ 16.12a	3.32 $\pm$ 0.16a	94.65 $\pm$ 12.80a
1	3.93 $\pm$ 1.34a	59.48 $\pm$ 1.86ab	58.33 $\pm$ 3.10a	0.061 $\pm$ 0.02a	138.61 $\pm$ 13.70b	3.12 $\pm$ 0.14a	120.87 $\pm$ 10.88ab
2	2.92 $\pm$ 0.95a	62.28 $\pm$ 1.54ab	64.27 $\pm$ 2.58a	0.045 $\pm$ 0.01a	185.69 $\pm$ 11.04bc	3.38 $\pm$ 0.12a	135.72 $\pm$ 9.05ab
3	4.15 $\pm$ 0.93a	63.12 $\pm$ 1.52ab	63.74 $\pm$ 2.53a	0.061 $\pm$ 0.01a	211.22 $\pm$ 11.19bc	3.35 $\pm$ 0.11a	148.03 $\pm$ 8.88b
4	5.49 $\pm$ 1.08a	66.25 $\pm$ 1.76b	60.70 $\pm$ 2.94a	0.074 $\pm$ 0.02a	227.05 $\pm$ 13.00c	3.34 $\pm$ 0.13a	157.89 $\pm$ 10.32b

Means followed by the same letter in any given column are not significantly different from one another ($\alpha = 0.05$) by Tukey's pairwise comparison test.

Table 3. Spearman correlation coefficients of the relationships among tomato quality measures

	Bruising level	Diam difference	Min. diam	Days to ripe	Roundness	No. of seeds	% sugar	Weight
Bruising level	1.00							
Diam difference	0.167	1.00						
Min. diameter	0.356**	0.247**	1.00					
Days to ripe	-0.060	-0.071	-0.052	1.00				
Roundness	0.132	0.991**	0.138	-0.064	1.00			
No. of seeds	0.623**	0.244*	0.682**	-0.197	0.170	1.00		
% sugar	0.016	-0.098	0.134	0.175	-0.109	0.038	1.00	
Weight	0.366**	0.435**	0.792**	0.351**	0.351	0.598**	0.117	1.00

**, $P < 0.01$ (two tailed); *, $P < 0.05$ (two-tailed).

diameter ($F = 3.72$; $df = 4, 99$; $P = 0.007$; $1-\beta = 0.871$), number of seeds ($F = 18.82$; $df = 4, 99$; $P < 0.0001$; $1-\beta = 1.00$), and tomato weight ($F = 4.67$; $df = 4, 99$; $P = 0.002$; $1-\beta = 0.941$) with respect to bruising level (Table 2). There was no difference among bruising levels with respect to the difference in diameter ($F = 1.59$; $df = 4, 99$; $P = 0.184$; $1-\beta = 0.474$), the number of days until the tomato was ripe ($F = 1.41$; $df = 4, 99$; $P = 0.235$; $1-\beta = 0.426$), tomato roundness ($F = 1.34$; $df = 4, 99$; $P = 0.259$; $1-\beta = 0.406$), or the percentage of sugars ($F = 0.60$; $df = 4, 99$; $P = 0.665$; $1-\beta = 0.191$). There was no increase in tomato weight, average diameter, or minimum diameter after a bruising level of 1, and there was no increase in the number of seeds after a bruising level of 2. Spearman's correlation showed that there were many significant relationships among the eight factors tested (Table 3). Number of seeds, weight, and minimum diameter were strongly correlated with bruising level (Figs. 4–6). Because bruising level was an ordinal data set, only linear analysis was appropriate; however, the relationship between bruising level and minimum diameter, number of seeds, and weight all appear to have a logarithmic relationship. Bruising level was most highly correlated with number of seeds ($r_s = 0.63$, $P < 0.0001$). The number of seeds was correlated with minimum average diameter ($r_s = 0.65$, $n = 108$, $P < 0.0001$; $y = 94.49 + 37.60\log x$; Fig. 7).

Based on the minimum diameter, the percentage of tomatoes that were commercial grade versus the number 2 grade was 7.9, 30.0, 42.9, 60.7, and 60.0% for bruising levels 0, 1, 2, 3, and 4, respectively (Fig. 8). Chi-squared analysis of the number of tomatoes in the commercial grade versus the number in the number 2 grade showed that there was a difference among bruising levels ($\chi^2 = 13.81$, $df = 4$, $P = 0.008$). When bruising level 0 was removed from the analysis, there was no difference in the number of tomatoes in the commercial grade versus the number 2 grade with respect to bruising level ($\chi^2 = 5.82$, $df = 3$, $P = 0.121$).

Discussion

Duration of bumble bee visits to tomato flowers was highly variable (unpublished data), and as a result, the number of bee visits to a flower may not be a good indicator of pollination level. Instead, bruising of the anther cone appeared to be a better indication of the contact duration of bumble bees with the stigma, and subsequent pollen transfer. We found that bruising of the anther cone was a good indicator of stigmatic pollen dose, and we recommend this method for assessing levels of bumble bee pollination of tomato flowers. Similarly, Jarlan et al. (1997) found that total contact duration of the drone fly, *Eristalis tenax* (L.), on sweet pepper flowers was a better indicator of pollen transfer

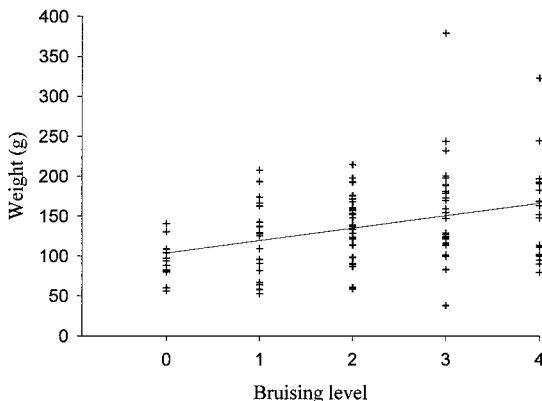


Fig. 4. Relationship between tomato weight and bruising level of tomato flower anther cones pollinated by bumble bees ($n = 108$; $r_s = 0.37$).

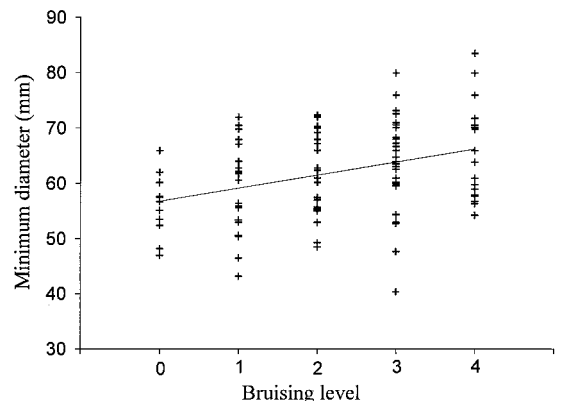


Fig. 5. Relationship between minimum tomato diameter and bruising level of tomato flower anther cones pollinated by bumble bees ($n = 108$; $r_s = 0.36$).

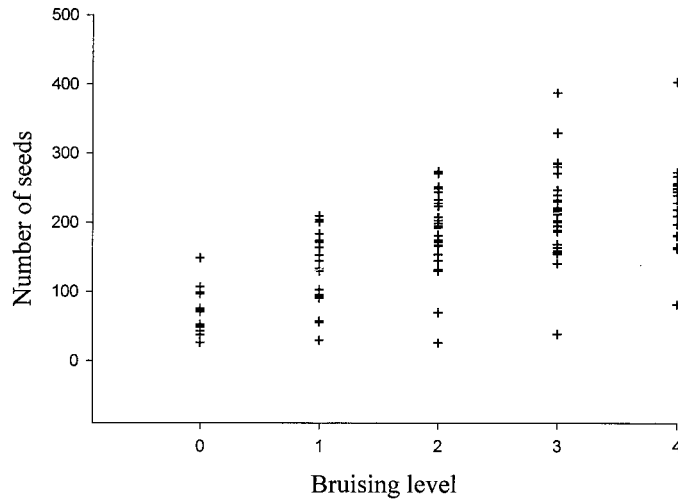


Fig. 6. Relationship between the number of seeds and bruising level of tomato flowers pollinated by bumble bees ($n = 108$).

to the stigma than the number of visits to the flower, which did not influence fruit characteristics.

Our results indicate that any level of bumble bee pollination increases fruit set in relation to no pollination. Similar results have been found in other studies comparing bumble bee pollination of tomato flowers to no pollination (Banda and Paxton 1991, van Ravestijn and van der Sande 1991, Kevan et al. 1991). In our study, bruising levels >0 did not result in significantly greater fruit set. Chen (1996) reported similar results for cantaloupe fruit set in relation to the number of honey bee visits. He found that fruit set was 3.1% in the absence of honey bees, and increased to 45, 69, and 59% for one, two, and three bee visits, respectively. His findings and ours indicate that there is a large difference in fruit set between pollinated and unpollinated flowers, but not between different levels of pollination. In our study, although there was no significant increase in fruit set between bruising levels,

there was a 17% increase in fruit set between the average fruit set of levels 1, 2, and 3 (83.0%) and level 4 (100%). However, the levels of fruit set within the experimental greenhouses may have been lower than what is experienced in the commercial greenhouses due to occasional high temperatures in the experimental greenhouses. In addition, bruising level 4 corresponds to the entire anther cone being bruised and damaged, and was never observed in the commercial greenhouses, indicating that realistic levels of pollination result in similar fruit set. Within commercial greenhouses, tomato clusters are pruned from seven to eight tomatoes usually down to four tomatoes, making quality measures more important than 100% fruit set.

It is well known that fruit characteristics such as seed set, size, and weight increase with greater levels of pollination (e.g., Chagnon et al. 1989, 1991; Chen 1996; Pressman et al. 1998). We also found positive trends between the level of pollination and the num-

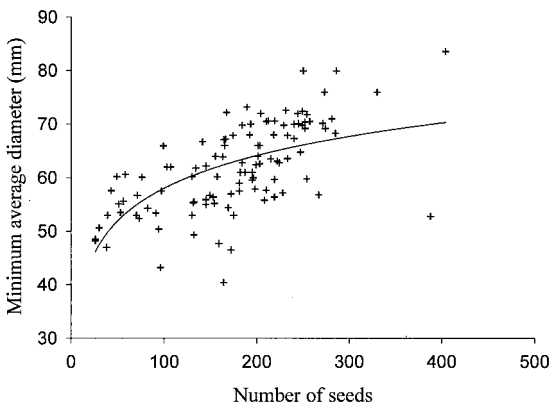


Fig. 7. Relationship between minimum diameter and the number of seeds ($\log_{10}$) of beefsteak tomatoes pollinated by bumble bees ($n = 108$; $y = 8.815 + 17.43\log x$).

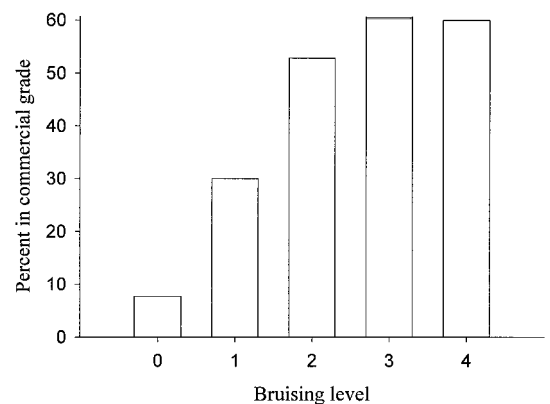


Fig. 8. Percent of tomatoes that were graded as commercial grade versus the lower quality number two grade based on a minimum diameter of 63 mm.

ber of seeds, minimum diameter, and weight, although we found diminishing return relationships. But, at what level is bruising and consequently tomato quality adequate? Although this is partly dependent on greenhouse growing conditions, and individual grower preferences, some conclusions and recommendations can be made from our study.

Of the variables that increased significantly with bruising level, minimum diameter is the most important when tomatoes are graded. The average minimum diameter increased only 0.84 mm (62.38–63.12 mm) from a bruising level of 2 to a bruising level of 3. The logarithmic relationship between minimum diameters with number of seeds indicated that greater levels of pollination result in lower returns in terms of size increases. Considering all factors that were significantly correlated with bruising level, we found that pollination of tomato flowers greater than a bruising level of 2 did not result in a significant increase in tomato quality. Although bruising levels >2 resulted in nonsignificantly larger tomatoes with more seeds, the increase was nonlinear, indicating diminishing returns with further pollination. Fruit quality based on our variables produced adequate quality tomatoes, based on the grades and standards outlined by the Canadian Food Inspection Agency.

Variation in some of the variables tested was high, resulting in low power in tests of differences among bruising levels. Specifically, the number of days for tomatoes to ripen, the percentage of sugars, and roundness were all tested at a power of <0.5. These variables should be tested again with larger sample sizes to test for bruising level effects.

Seed count was probably the most accurate indicator of sufficient pollination in our study, because other quality measures were influenced by environmental conditions such as plant resources and greenhouse conditions (Picken 1984). Because no increase in fruit set, weight, and diameter were detected after a bruising level of 1, and no significant increase in the number of seeds after a bruising level of 2, we recommend a bruising level of 1–2 as adequate to ensure sufficient fruit set and quality of greenhouse tomatoes. Although in this experiment, tomatoes were grown in a small experimental greenhouse, the relationship found between bruising level and quality should be similar under commercial growing conditions. In Southern Ontario, growers and bee suppliers have been aiming for heavy bruising on the anther cones. This study indicates that intense bruising may not be required to ensure adequate pollination and consequently, colony densities may not need to be as high as previously thought.

Acknowledgments

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